

Acme Engineering Handbook

Platform Engineering Standards — Edition 2024

1. Architecture Overview

The Acme platform is a microservices architecture deployed on EKS across three AWS regions (us-east-1, eu-west-1, ap-south-1). Services communicate via gRPC for internal calls and GraphQL for external APIs. The primary datastore is PostgreSQL 15 with read replicas in each region. Redis handles caching, and Kafka (MSK) handles event streaming. Object storage uses S3 with cross-region replication and Object Lock for compliance data.

All services must expose health endpoints at `/healthz` and `/readyz`, a Prometheus metrics endpoint at `/metrics`, and an OpenTelemetry trace exporter. Service discovery uses Kubernetes native DNS plus a service mesh (Istio) for mutual TLS and traffic shaping.

2. Deployment Process

All deployments go through GitHub Actions with reusable workflows. Production deploys require two approvals and must occur between 09:00 and 17:00 in the deploying team's primary timezone. The deployment freeze window is the last Friday of each month and the final two weeks of December. Rollback must be possible within 3 minutes.

All production deployments use progressive delivery with Argo Rollouts. Canary analysis is mandatory for tier-1 services. Automated rollback triggers on SLO burn-rate alerts. Feature flags are managed through LaunchDarkly and must be removed within 30 days of full rollout.

3. On-Call Rotation

Each team maintains a primary and secondary on-call rotation with weekly shifts. PagerDuty is the source of truth for escalation. Sev-1 incidents require a response within 10 minutes; Sev-2 within 30 minutes; Sev-3 within 4 hours. Postmortems are mandatory for all Sev-1 and Sev-2 incidents and must be filed within 3 business days.

On-call compensation remains \$350 per week. Comp time is provided at 1 hour per 3 hours of after-hours work. A new follow-the-sun rotation is available for tier-1 services with teams in at least two timezones.

A new incident command training program is required for all engineers before joining an on-call rotation. Training covers incident classification, communication protocols, and postmortem facilitation.

4. Code Review Standards

Every pull request requires at least one approving review from a code owner. PRs touching authentication, payments, data deletion, or personally identifiable information require two approvals. Maximum PR size is 400 lines of diff; larger changes must be split into stacked PRs. All PRs must pass linting, unit tests, integration tests, and security scans before merge.

A new AI-assisted review step uses a static analysis tool trained on the Acme codebase to flag common issues before human review. Automated suggestions do not replace human reviewers.

Architectural review is now asynchronous. Proposals are submitted as ADRs (Architecture Decision Records) in the docs repository and reviewed by the Architecture Guild within 5 business days.

5. Observability

All services emit structured JSON logs to the central ELK stack with a 30-day retention in hot storage and 12-month retention in cold storage. Metrics are scraped by Prometheus every 15 seconds with a 15-month retention. Distributed traces are captured via OpenTelemetry with a 10 percent sampling rate in production and 100 percent in staging.

Service Level Objectives (SLOs) are defined for all tier-1 and tier-2 services. Error budgets are tracked weekly. Alerts are tied to SLO burn rates. Every alert must have a runbook and a designated owner; orphan alerts are auto-disabled after 30 days.

6. Testing Standards

Unit test coverage must be at least 85 percent per service. Integration tests must cover all public endpoints and at least 90 percent of internal service boundaries. Contract tests are required for all service-to-service dependencies. End-to-end tests run on every merge to main and nightly on staging.

Load testing is performed quarterly for tier-1 services. Chaos engineering experiments run monthly in staging and quarterly in production during business hours with a canary-only blast radius.

7. Security

All production secrets are stored in AWS Secrets Manager with automatic rotation every 90 days. Static analysis (SAST), dynamic analysis (DAST), and dependency scanning run on every PR. Container images are scanned with Trivy before promotion. Software bills of materials (SBOMs) are generated for every release.

A new zero-trust network model is being rolled out. All service-to-service traffic uses mutual TLS. Access to production systems requires hardware-key MFA and just-in-time elevation.

8. Data Engineering

Batch pipelines use Apache Airflow with a shared DAG repository. Streaming pipelines use Kafka Streams and Flink. The data lake is built on S3 with a Glue catalog and queried via Athena and Snowflake. All pipelines must have data quality checks and lineage tracking via OpenLineage.

9. Machine Learning Platform

Model training runs on SageMaker and a self-hosted Ray cluster. Model registry is MLflow. Feature store is Feast. Online inference is served via KServe on EKS. All models must have a documented model card, an evaluation report, and a rollback plan before promotion to production.